

RESEARCH AND DEVELOPMENT ENGINEER

CFD engineer with 4 years of research experience in multiphase, turbulent, and compressible flow simulation, spanning both high-fidelity academic research and industrial application. Developed interface-capturing and CFD-DEM methods for CEA on the FUGU and YALES2 codes; validated aerodynamic simulations with close agreement to S2A wind tunnel data at Stellantis. Fluent in Fortran, Python, and C across HPC environments (MPI).

PROFESSIONAL AND ACADEMIC EXPERIENCE

- 2025 **Research Engineer** [*Université Gustave Eiffel/ESYCOM*](#)
- Designed a **CFD-DEM** simulation pipeline for air quality sensor modeling on the YALES2 HPC code,
 - Validated particle-laden flow models on YALES2 across 3 configurations (fluidized bed, pipe fouling, inertial impaction),
 - Optimized collision modeling and drag force correlations in **CFD-DEM** for dilute regimes; extended particle/fluid interaction models (turbulence, drag, lubrication),
 - Implemented subgrid-scale (**SGS**) turbulent dispersion model for **LES**-coupled **CFD-DEM**.
- 2021 - 2024 **PhD Researcher** [*Université Gustave Eiffel*](#)
- Developed and implemented resolved interface capturing methods in the **FUGU** in-house code for **CEA** (benchmarked against **Ansys Fluent**),
 - Characterized splashing mechanisms in liquid pool impacts under compressible conditions using **VOF** method,
 - Quantified **Rayleigh-Plateau instability** growth rates to study jet fragmentation and satellite droplet dynamics,
 - Built automated **Python/ParaView** pipeline for detection and counting of resuspended particles across simulation batches.
- 2021 - 2024 **Part-time Lecturer in Thermodynamics** [*Université Gustave Eiffel*](#)
- Delivered practical sessions in numerical simulation on **Ansys Fluent**,
 - Taught thermodynamics and heat transfer (lectures and tutorials),
 - Taught thermal machines and heat pump systems (lectures and tutorials).
- 2021 **Aerodynamic Simulation Engineer** [*Groupe Stellantis*](#)
- Performed RANS aerodynamic simulation (k- ϵ Realizable) of the Citroën DS7; validated against **S2A** wind tunnel data (PIV + force balance), achieving close numerical/experimental agreement on drag and lift,
 - Conducted parametric study of 5 rear wheel configurations (rim porosity, track width ± 6 mm) and quantified their impact on drag and lift torsors,
 - Designed and simulated 3 air duct geometries in the wheel arch, achieving local drag reduction at the wheelhouse.
- 2018 **Aeronautical Technician** [*Air Algérie*](#)
- Conducted maintenance inspections on turbojet and turboprop engines.
 - Validated post-maintenance engine performance on full-scale test bench (1:1).

EDUCATION

- 2021 - 2024 **PhD in Fluid Mechanics** [*Université Gustave Eiffel*](#)
- Modeling and simulation of droplet and spray impacts on liquid pools.
Simulation with the **FUGU** code developed at the **MSME** laboratory.
- 2019 - 2021 **MSc in Mechanics** [*Université Gustave Eiffel*](#)
- Modeling and Simulation in Fluid Mechanics and Heat Transfer.
Courses : Fluid mechanics, heat transfer, turbulence, multiphase flows, porous media, numerical methods, fluid and solid numerical simulation (*Ansys Fluent/Comsol Multiphysics*), acoustics.
- 2015 - 2019 **MSc in Aeronautical Sciences** [*IAES, Université de Blida*](#)
- Bachelor's and Master's 1 in aircraft propulsion.
Courses : Fluid mechanics, heat transfer, thermodynamics (*Internal combustion engines*), numerical methods, scientific programming (*Fortran*), numerical simulation (*Ansys Fluent*), compressible flows.

ADDITIONAL ACTIVITIES

- Activities Manager** [*Association MSFT*](#)
- Organized events and hands-on training workshops (Ansys Fluent, Programming, CV writing) for engineering students.
- Workshop Facilitator** [*Association MSFT*](#)

Led Python scripting and MPI parallel programming (Fortran) workshops.

PhD Student Representative at ED-SIE Doctoral School

[*École doctorale ED-SIE*](#)

Organized networking days and scientific events for the doctoral school. Liaison between PhD students and the doctoral school management.